

MAIKEISE'S WARE

DOMAIN-DRIVEN DESIGN DOCUMENT (DDD)

DDD - Quotation Management & Search System

Technical pre-project, architecture, domain model, data governance and evolution roadmap.

Anny Maikese

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Version 1.0 | Google Apps Script + Google Sheets

Status: Initial baseline approved for development start.

1. Document control

Field	Description
Organization	Maikese's Ware
Owner / codename	Anny Maikese
Product	Quotation Management & Search System
Version	1.0
Baseline date	2026-09-16
Technology	Google Apps Script + Google Sheets

1.1 Purpose

Formalize the pre-project, domain, architectural decisions and evolution baseline. This document is the technical baseline for version 1.0.

2. Product vision and scope

The system will centralize a historical quotation archive into a structured and searchable repository. Version 1 focuses on two core capabilities: registering and searching quotations.

2.1 Business goals

- Centralize and standardize historical data.
- Reduce information retrieval time.
- Support optional complementary information.
- Preserve traceability and integrity.
- Operate at low cost within the Google ecosystem.

2.2 Out of scope for V1

- Automatic OCR.
- ERP and external APIs.
- Approval workflow.
- Native mobile application.
- SQL database.

3. DDD - domain and ubiquitous language

Core domain: Quotation Management.

Domain	Business definition
Quotation	Commercial record associated with an RFP and item.
RFP	Quotation/request process identifier.
Item	Quoted product, material or service.
Supplier	Company/source responsible for the offer.
Optional data	Non-mandatory quotation complements.

3.1 Bounded Context

V1 has one bounded context: Quotations. OCR, documents, suppliers and ERP integrations should become separate modules/contexts in future versions.

3.2 Aggregate

Quotation is the aggregate root and may only be persisted after its invariants are validated.

4. Domain model

Field	Mandatory	Technical type	Technical rules
RFP	Yes	Text	Preserve formatting.
Item code	Yes	Text	Preserve leading zeros.
Reference	Yes	Text	Technical/commercial reference.
Description	Yes	Long text	Full item description.
Supplier	Yes	Text	Provided name.
Quantity	Yes	Decimal	> 0.
Unit value	Yes	BRL	>= 0.
Total value	Yes	BRL	Quantity x unit value.
NCM	Yes	Text	Persist as text.
Lead time	Yes	Text	Provided commercial condition.
Payment	Yes	Text	Provided condition.
Delivery	Yes	Text	Condition/location/mode.
Requested date	Yes	Date	Valid date.
Optional data	No	Key/value	Zero or many.

4.1 Technical metadata

Each record includes a unique ID, status, creation timestamp, update timestamp and version for concurrency control.

4.2 Optional data

Additional information is stored as key/value pairs. Examples include owner, currency, region, cost center, notes, taxes and other commercial conditions. Recurring fields may later be promoted to the main schema.

5. Solution architecture

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HTML/CSS/JavaScript
↓ google.script.run
Application Services
↓
Domain
↓
Repositories / SpreadsheetApp
↓
Google Sheets

```

Layer	Responsibility
UI	Registration, search, feedback and pagination.
Application	Orchestrates use cases.
Domain	Validation, calculations and business rules.
Repository	Maps entities to spreadsheet structures.
Infrastructure	Google services and configuration.

6. Persistence model

A dedicated spreadsheet is the source of truth. Direct spreadsheet editing should be restricted.

Entity	Properties
COTACOES	Mandatory fields, metadata and status.

DADOS_OPCIONAIS	Key/value pairs linked by quotation ID.
AUDITORIA	CREATE, EDIT and DEACTIVATE events.
CONFIG	Schema version and non-sensitive parameters.

Sensitive identifiers and configuration should preferably use PropertiesService.

7. Use cases

7.1 Registration screen

- Display all 13 mandatory business fields.
- Calculate total value automatically.
- Allow multiple optional key/value fields.
- Validate client-side and server-side.
- Generate ID, timestamps, status and version.

7.2 Search screen

- Search by RFP, item code, reference, description, supplier and NCM.
- Filter by supplier and date.
- Paginate results.
- Open full details.
- Edit/deactivate according to permissions.

8. Business rules

BR-001	All 13 business fields are mandatory.
BR-002	Quantity must be greater than zero.
BR-003	Unit value cannot be negative.
BR-004	Total value is calculated by the system.
BR-005	RFP, item code and NCM are stored as text.
BR-006	Requested date must be valid.
BR-007	Optional data may contain zero or many entries.
BR-008	Deactivation is preferred over physical deletion.
BR-009	Updates increment the record version.
BR-010	Writes must use locking.

A possible duplicate based on RFP + Item Code + Supplier should initially trigger a non-blocking warning.

9. Non-functional requirements and security

- Batch spreadsheet operations.
- Server-side pagination.
- Server-side validation.
- LockService for concurrency.
- Least-privilege permissions.
- Change auditing.
- No passwords/tokens in cells or GitHub.

- Backups according to archive criticality.

The solution is subject to Google Apps Script quotas and limits. Significant growth in volume or concurrency is an architectural review trigger.

10. ADR-001 - Initial stack

Context	Decision
Context	Internal system, low cost and rapid deployment.
Decision	Google Apps Script + Google Sheets.
Benefits	Low infrastructure overhead and native integration.
Risks	Quotas, concurrency and performance at scale.
Review when	Volume, SLA or integrations demand it.

11. Quality and acceptance criteria

- Register a valid quotation.
- Block missing mandatory fields.
- Calculate totals correctly.
- Store optional data without changing the main schema.
- Search by key fields.
- Open complete details.
- Edit while preserving ID and incrementing version.
- Deactivate without deleting history.
- Audit changes.

12. Roadmap

Version	Features
V1	Registration, search, detail, edit, deactivation and audit.
V1.1	Filters, controlled export and operational improvements.
V2	Document association and assisted OCR with human review.
V3	Dashboards, price history and supplier comparison.
V4	ERP/API integrations.
Future scale	Migrate persistence while preserving domain contracts.

13. Evolution governance

Every relevant change must create a new DDD version. Architectural decisions must receive an ADR containing context, alternatives, decision, consequences, impact, migration, date and related version.

13.1 Version history

Version	Date	Description	Author
1.0	2026-09-16	Initial project baseline.	Anny Maikese Maikese's Ware

14. Baseline approval

Role	Signature	Date
Technical owner	Anny Maikese	2026-09-16

Role	Name / Signature	Date
Product owner		
Key user / archive owner		

15. Technical references

Official Google documentation to be revisited before deployment changes: Apps Script Web Apps, Spreadsheet Service, Authorization and Quotas.

developers.google.com/apps-script/guides/web

developers.google.com/apps-script/reference/spreadsheet

developers.google.com/apps-script/guides/services/authorization

developers.google.com/apps-script/guides/services/quotas

END OF DOCUMENT

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